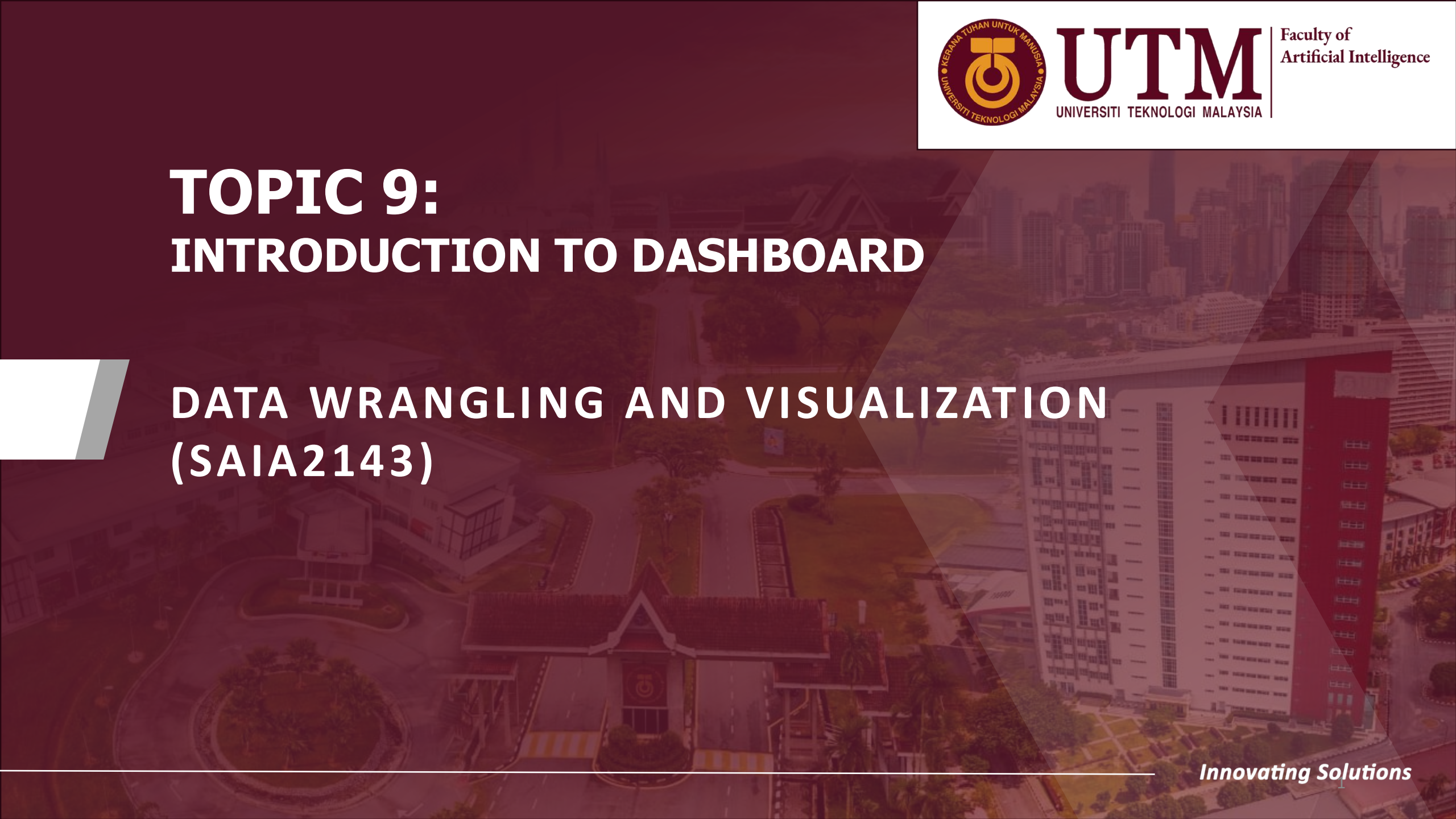


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Faculty of Artificial Intelligence

TOPIC 9:
INTRODUCTION TO DASHBOARD

DATA WRANGLING AND VISUALIZATION (SAIA2143)

COURSE LEARNING OUTCOMES

At the end of this topic, students should be able to:

a. Explain the concept and role of visualization dashboards in data analysis and decision-making processes.

b. Differentiate between various types of dashboards (e.g., operational, strategic, analytical, and tactical) and identify their appropriate use cases.

c. Describe the key components of a data dashboard, including metrics, data sources, interactivity, and layout structures.

d. Analyze the purpose and audience of a dashboard to guide the selection of content and visualization types.

e. Apply dashboard design principles to create clear, user-friendly, and impactful data visualizations.

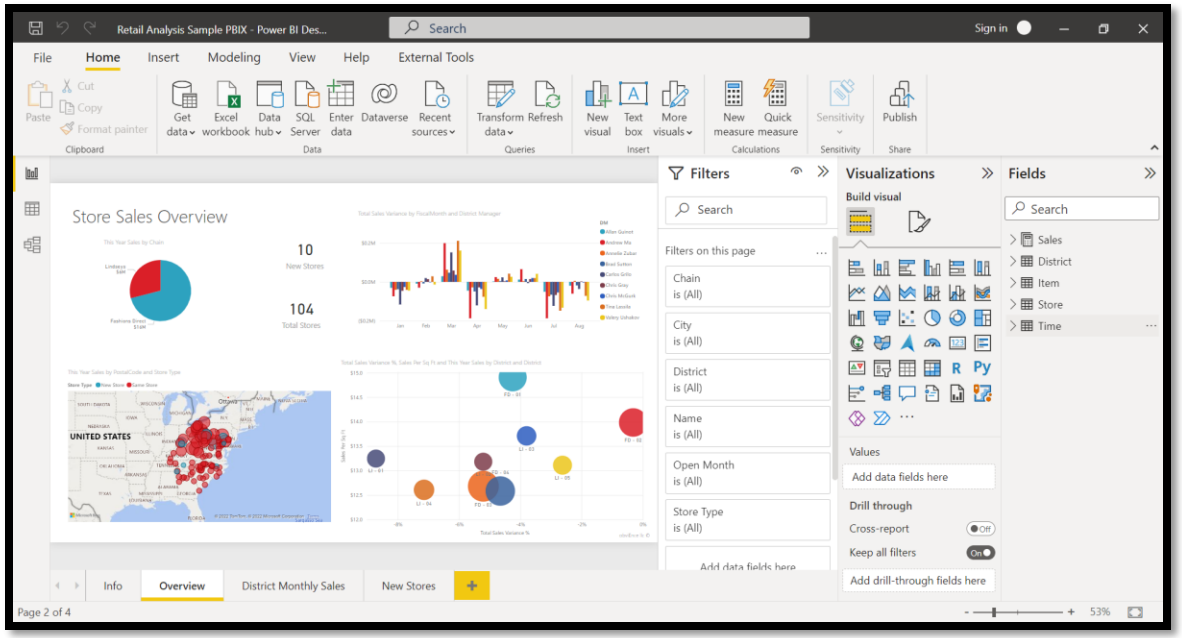
f. Build and configure a basic interactive dashboard using Google Looker Studio.

TOPIC 9: INTRODUCTION TO DASHBOARD

a. **Introduction to Visualization Dashboard**
b. Understanding the purpose and components of data dashboards
c. Design principles for effective dashboards

What is a Dashboard?

- **Definition**: A dashboard is a visual interface that provides at-a-glance views of key performance indicators (KPIs) relevant to a particular objective or business process.
- **Purpose**:
  i. Monitor metrics
  ii. Track performance
  iii. Enable data-driven decision making



Why Dashboard?

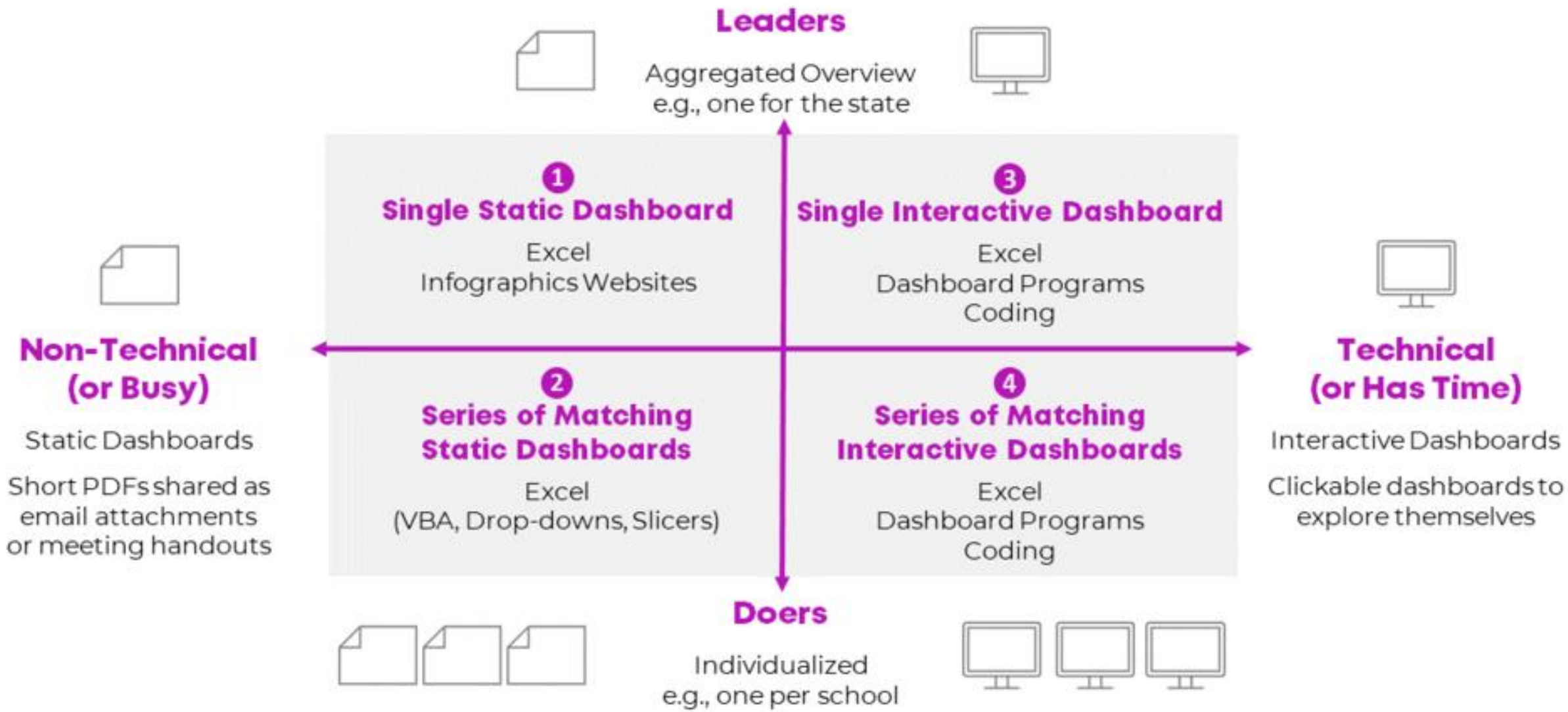
- Dashboards are decision-making tools.
- Poor design leads to confusion and misinterpretation.
- Good design improves usability, clarity, and trust in data.
- User-centered design = better engagement and outcomes.

Know Your Audience

- Different users need different data:
  a. Executives: Strategic overviews
  b. Analysts: Detailed exploration
  c. Managers: Operational KPIs

- Tailor design, complexity, and content accordingly

What Type of Dashboard Do We Need?





CLASS ACTIVITY

According to the YouTube link (11:52min): https://www.youtube.com/watch?v=Hf57j2xcR0g&t=2s

1. Determine the most suitable **dashboard type** for the scenario:
    a. **Static vs. Interactive**
    b. **Single-page vs. Series**

2. Justify the choice based on:
    a. **Audience Needs**
    b. **Data Complexity**
    c. **Update Frequency**

Scenario-Based Group Activity

1. Executive Summary:

A CEO needs a high-level overview of quarterly performance metrics.

2. Marketing Campaign Analysis:

A marketing team wants to track real-time engagement metrics during a new product launch.

3. Financial Reporting:

The finance department requires monthly reports for budget tracking.

4. Customer Support Monitoring:

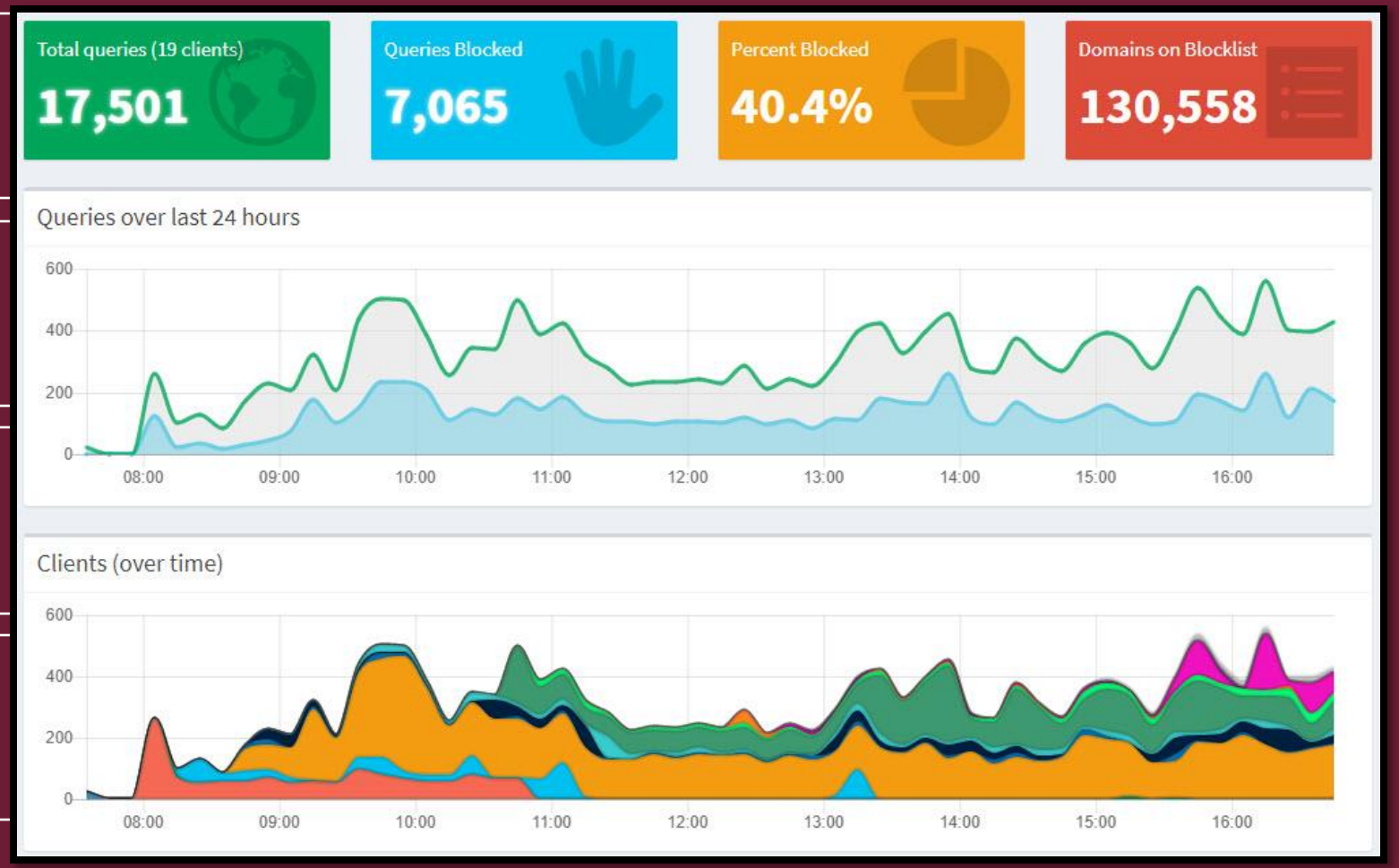
A support manager needs to monitor daily ticket resolutions and response times.

ANSWER

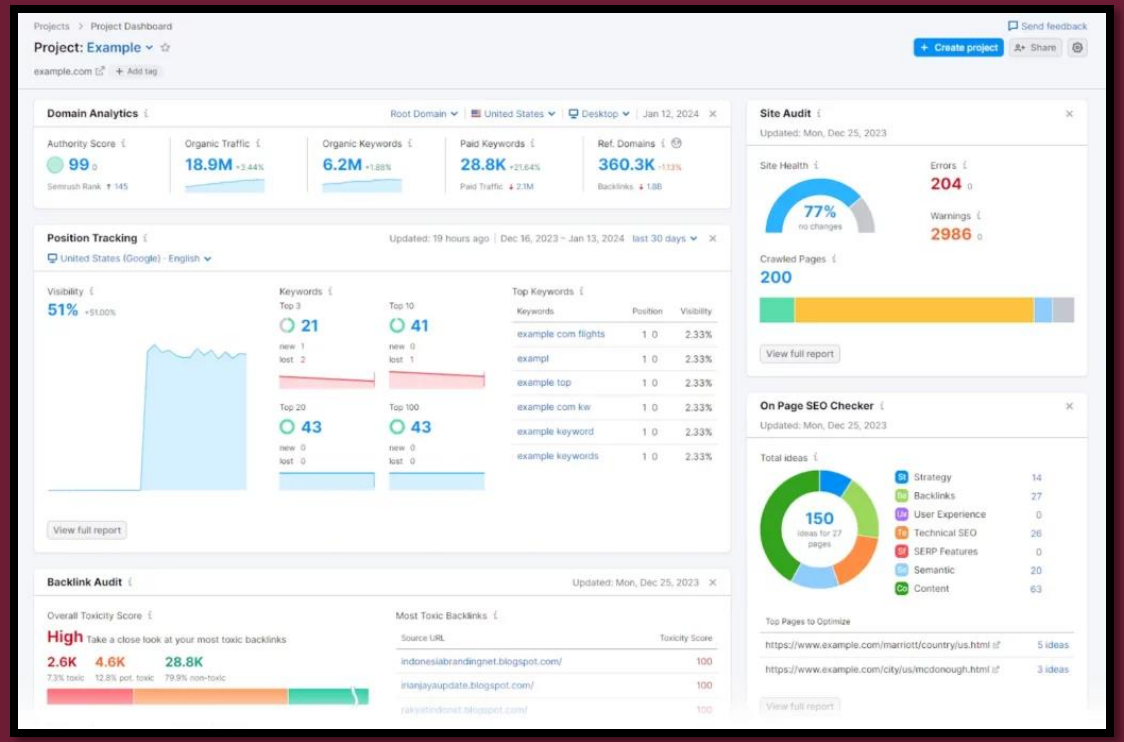
Scenario	Interactivity	Layout Type	Reasoning Highlights
Executive Summary			
Marketing Campaign Analysis			
Financial Reporting			
Customer Support Monitoring			

Type of Dashboards

1. Analytical
2. Operational
3. Strategic
4. Tactical



1. Analytical Dashboards



- **Purpose**: To identify trends, patterns, and insights over time, supporting strategic decision-making.
- **Key Features**:
  a. Detailed visualizations such as line graphs, scatter plots, and histograms.
  b. Focus on historical data analysis.
  c. Used by data analysts or business intelligence teams.
- **Example**: A sales performance dashboard analyzing historical sales data to identify top-performing products and seasonal trends.

2. Operational Dashboards



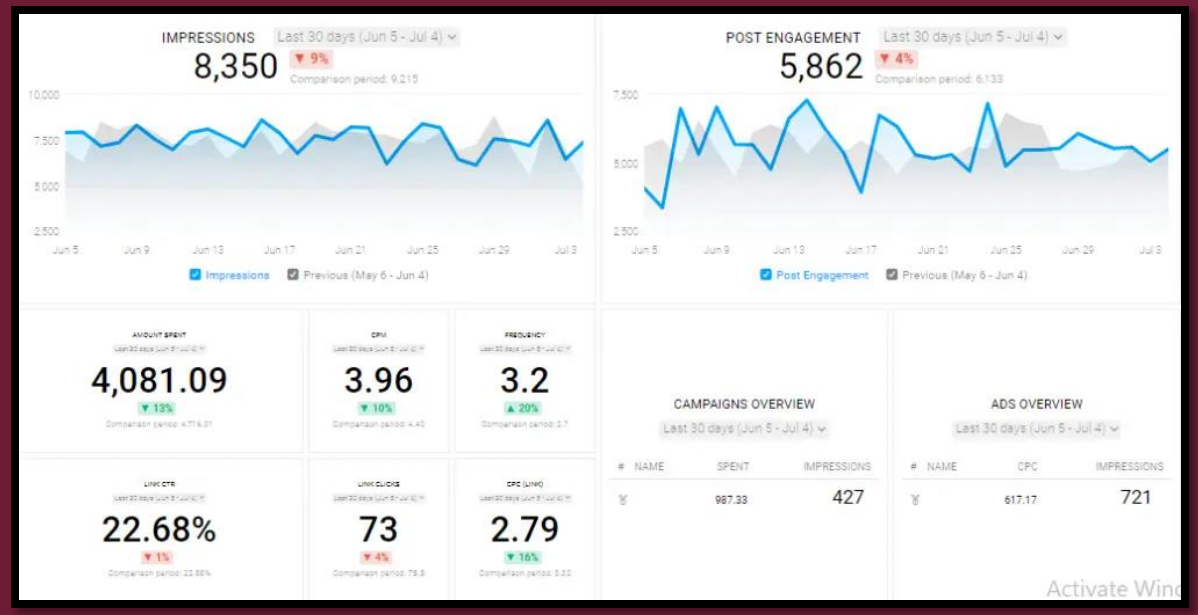
- **Purpose**: For real-time monitoring and quick decision-making in daily operations.
- **Key Features**:
  a. Displays frequently updated data like daily sales figures or website traffic.
  b. Emphasizes real-time data visualization.
  c. Ideal for managers tracking day-to-day operations.
- **Example**: A customer support dashboard monitoring open tickets, average response time, and agent availability in real time.

3. Strategic Dashboards



- **Purpose**: To provide a high-level overview of organizational performance, focusing on long-term goals.
- **Key Features**:
  a. Summarizes key performance indicators (KPIs) like revenue growth or customer satisfaction.
  b. Prioritizes simplicity and clarity.
  c. Used by executives for strategic decision-making.
- **Example**: An executive KPI dashboard tracking quarterly revenue, year-to-date profit margins, and employee engagement scores.

4. Tactical Dashboards



- **Purpose**: To bridge the gap between operational and strategic dashboards, focusing on short-term goals.
- **Key Features**:
    a. Provides insights into specific areas like project management or marketing campaign performance.
    b. Used by mid-level managers for tactical decision-making.
- **Example**: A marketing campaign dashboard tracking metrics such as click-through rates, conversion rates, and ROI for ongoing campaigns.

Dashboard vs. Report

Feature	Dashboard	Report
Format	Visual, interactive	Text-heavy, static
Usage	Monitoring, quick insights	Documentation, analysis
Frequency	Real-time or frequent	Periodic

Dashboards and reports are complementary but not the same.

TOPIC 9: INTRODUCTION TO DASHBOARD

a. Introduction to Visualization Dashboard
b. Understanding the purpose and components of data dashboards
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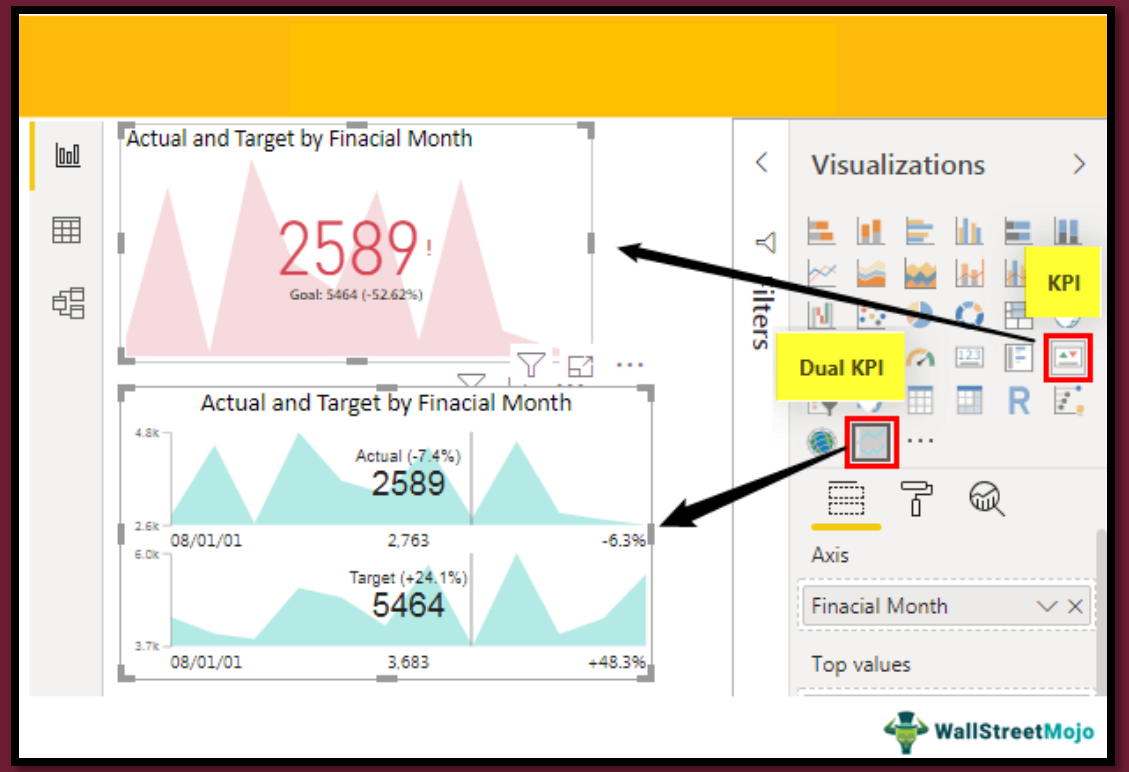
Purpose of Data Dashboards

a. Real-time monitoring of business activities
b. Decision support tool for management and teams
c. Provide insights into performance and trends
d. Reduces time to insight
e. Improves transparency and accountability
f. Enhances communication with stakeholders

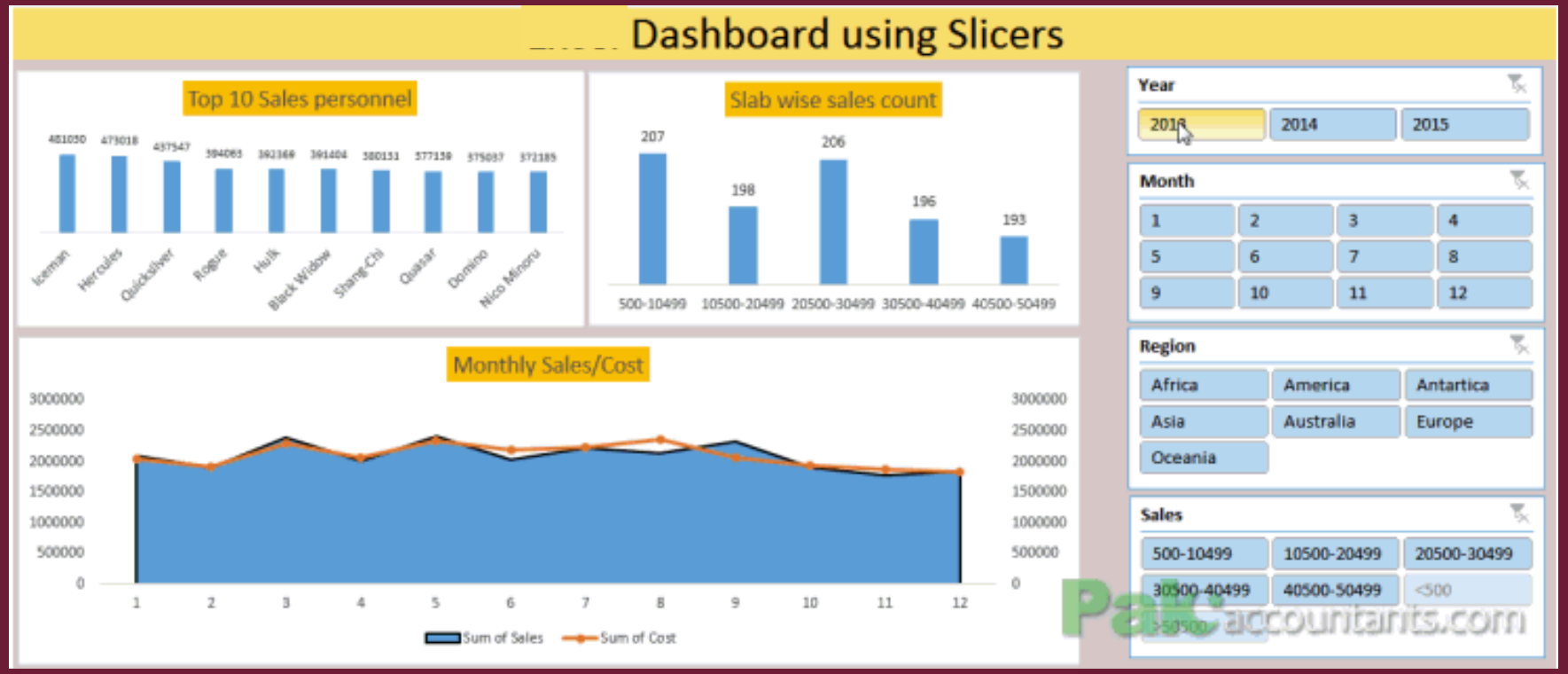
Core Components of a Dashboard

a. **KPIs & Metrics**: Key numbers tracked against goals
b. **Visualizations**: Charts, graphs, maps, gauges





Core Components of a Dashboard



c. **Filters & Slicers**: Allow user interaction and customization

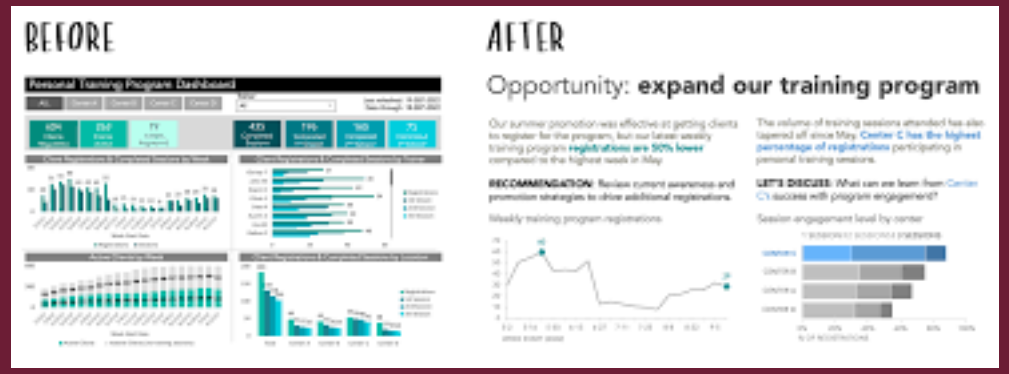
d. **Data Sources**: Connected systems (e.g., Excel, SQL, cloud data)

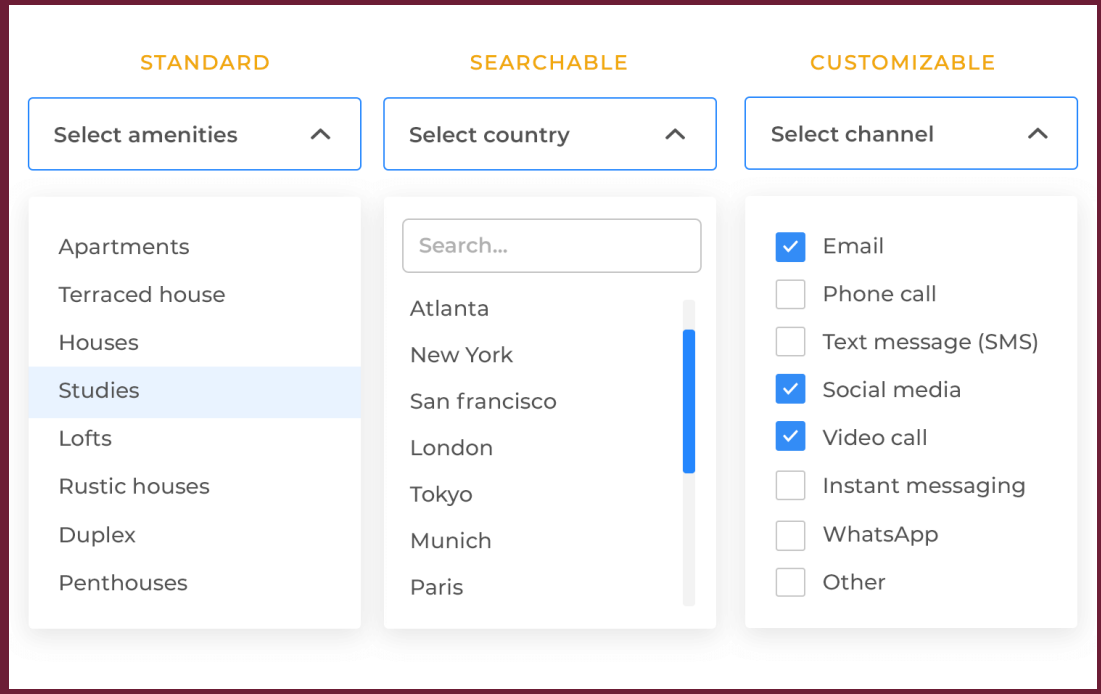
www.fai.utm.my

Core Components of a Dashboard

e. **Navigation Controls**: Tabs, buttons, drop-downs

f. **Textual Context**: Titles, descriptions, data notes

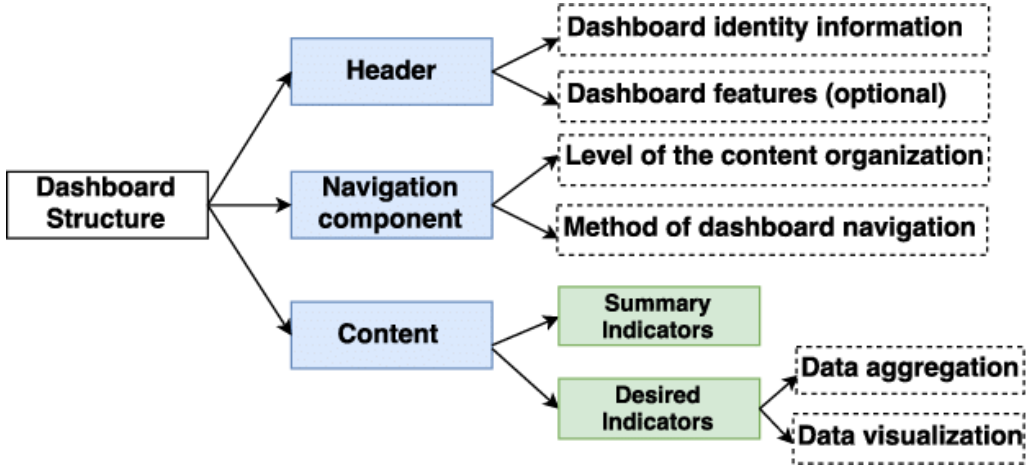




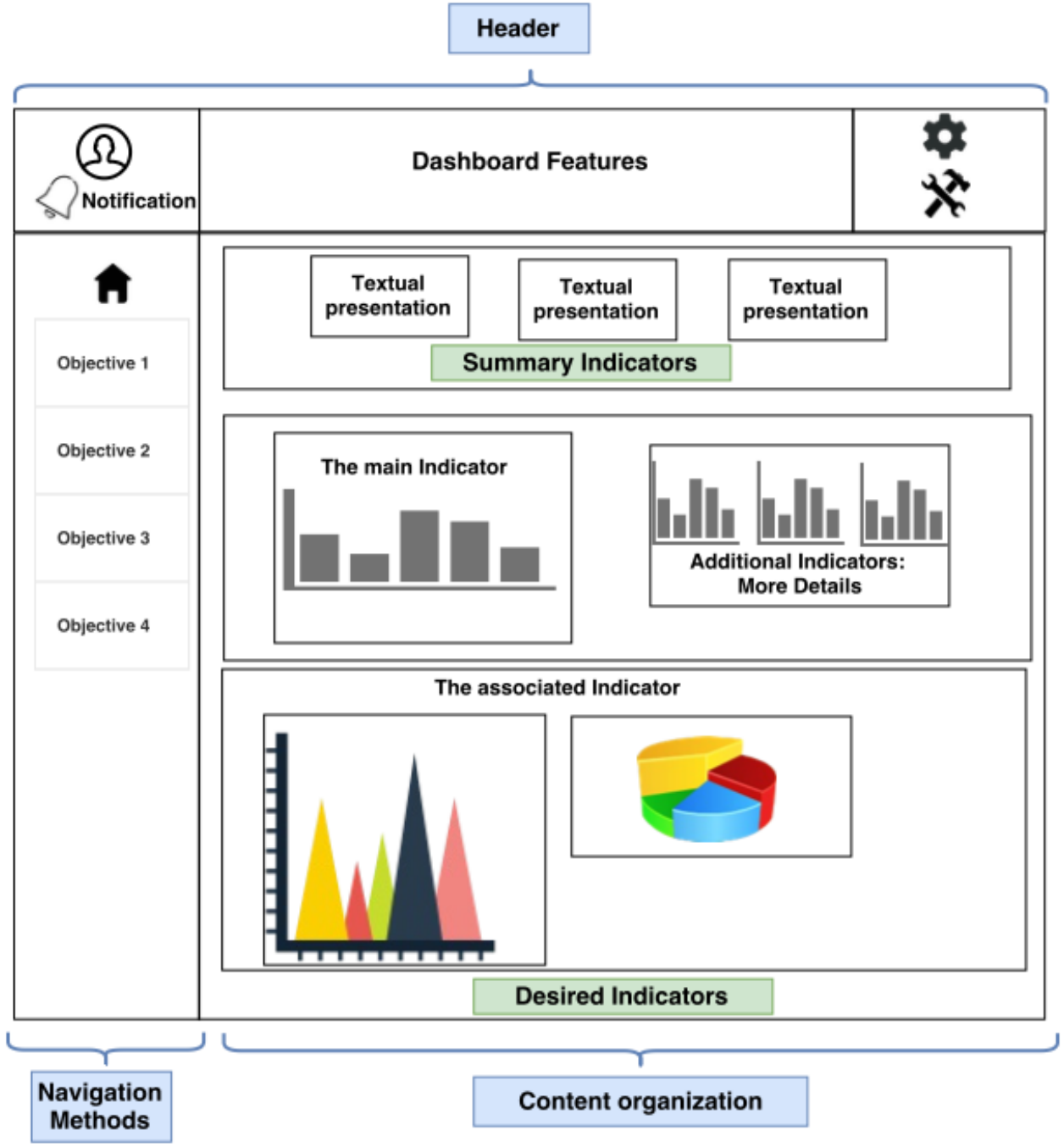
Best Practices for Using Components

- Keep it simple and focused
- Use consistent design patterns
- Prioritize the most relevant KPIs
- Limit number of filters to avoid confusion
- Validate data source accuracy

Example in Research



Ines, D., et al. (2017).



CLASS ACTIVITY



Please identify real-world examples of dashboards:

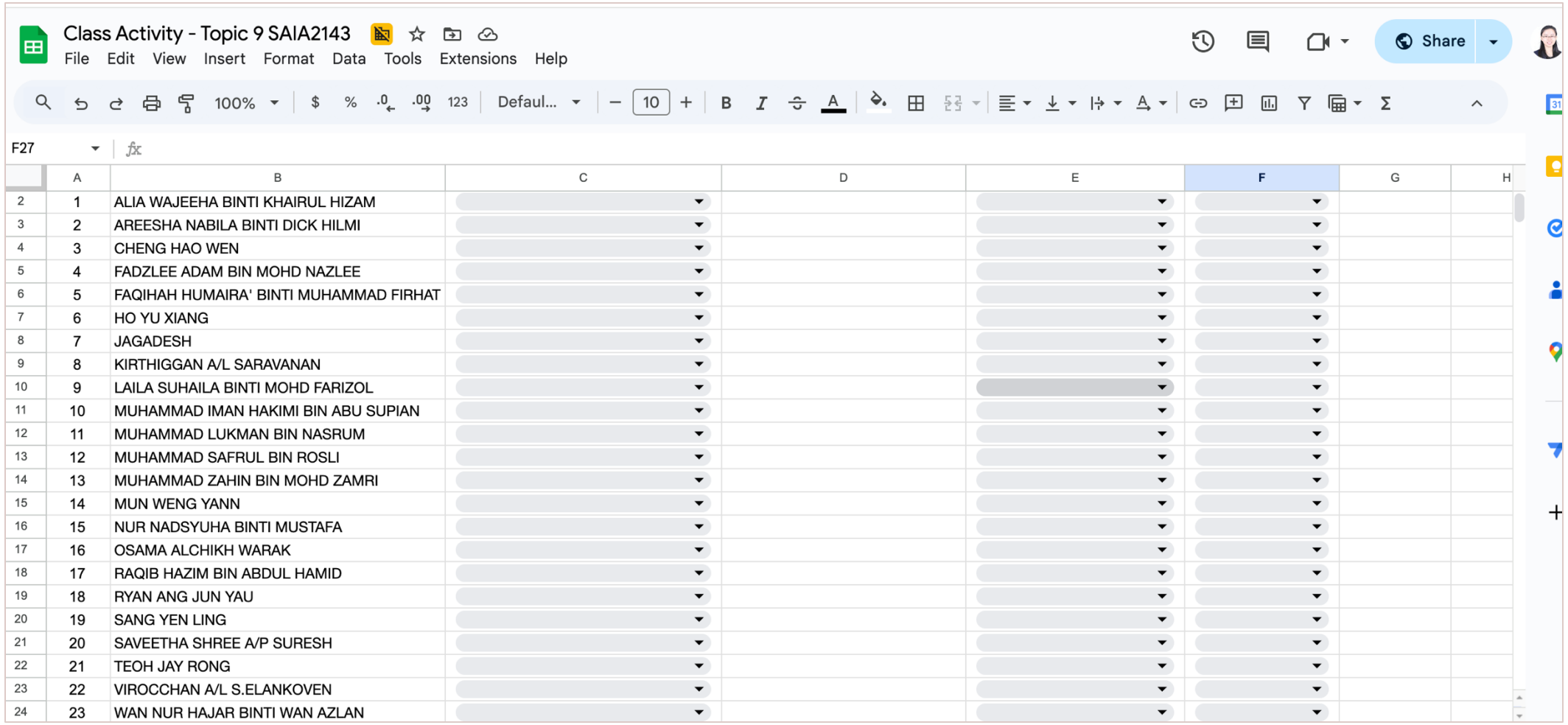
a. Business
b. Healthcare
c. Education

What are the example, website link, type, component you see inside the dashboards?

Answer in Google Sheets.

https://docs.google.com/spreadsheets/d/13IxBa4Ptf_jrDUb_reAHIfurTQ3PmSdLf1komzYC7fw/edit?usp=sharing

https://docs.google.com/spreadsheets/d/13IxBa4Ptf_jrDUb_reAHIfurTQ3PmSdLf1komzYC7fw/edit?usp=sharing



	A	B	C	D	E	F	G	H
2	1	ALIA WAJEEHA BINTI KHAIRUL HIZAM						
3	2	AREESHA NABILA BINTI DICK HILMI						
4	3	CHENG HAO WEN						
5	4	FADZLEE ADAM BIN MOHD NAZLEE						
6	5	FAQIHAH HUMAIRA' BINTI MUHAMMAD FIRHAT						
7	6	HO YU XIANG						
8	7	JAGADESH						
9	8	KIRTHIGGAN A/L SARAVANAN						
10	9	LAILA SUHAILA BINTI MOHD FARIZOL						
11	10	MUHAMMAD IMAN HAKIMI BIN ABU SUPIAN						
12	11	MUHAMMAD LUKMAN BIN NASRUM						
13	12	MUHAMMAD SAFRUL BIN ROSLI						
14	13	MUHAMMAD ZAHIN BIN MOHD ZAMRI						
15	14	MUN WENG YANN						
16	15	NUR NADSYUHA BINTI MUSTAFA						
17	16	OSAMA ALCHIKH WARAK						
18	17	RAQIB HAZIM BIN ABDUL HAMID						
19	18	RYAN ANG JUN YAU						
20	19	SANG YEN LING						
21	20	SAVEETHA SHREE A/P SURESH						
22	21	TEOH JAY RONG						
23	22	VIROCCHAN A/L S.ELANKOVEN						
24	23	WAN NUR HAJAR BINTI WAN AZLAN						

TOPIC 9: INTRODUCTION TO DASHBOARD

a. Introduction to Visualization Dashboard
b. Understanding the purpose and components of data dashboards
c. **Design principles for effective dashboards**

CLASS ACTIVITY

Instruction:

According to the YouTube video please list down **12 dashboard design** tips for better data visualization that you can apply in design.



YouTube link (9:50min):
https://www.youtube.com/watch?v=t3cAUt7sOQg

Importance Of Dashboard UI & UX Design Principles

1. To keep audiences engaged
2. To convey information correctly
3. To identify patterns and trends

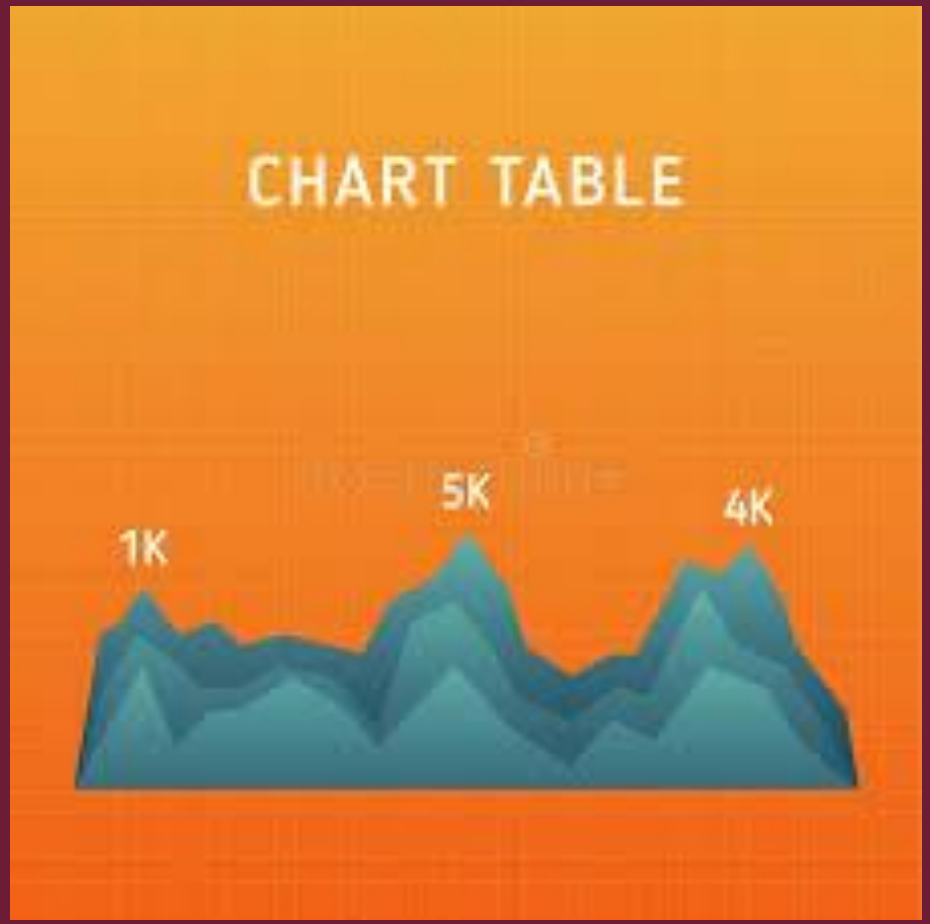
Focus on the Most Important Data

- Prioritize critical KPIs
- Avoid clutter and information overload
- Highlight trends, alerts, and deviations
- Use layout to emphasize priority metrics

Choose the Right Visualizations

- Bar/column charts: Compare values
- Line charts: Show trends over time
- Pie charts: Show parts of a whole (use sparingly)
- Gauges, cards: Highlight KPIs

- **Avoid chart junk:** 3D, overly stylized visuals

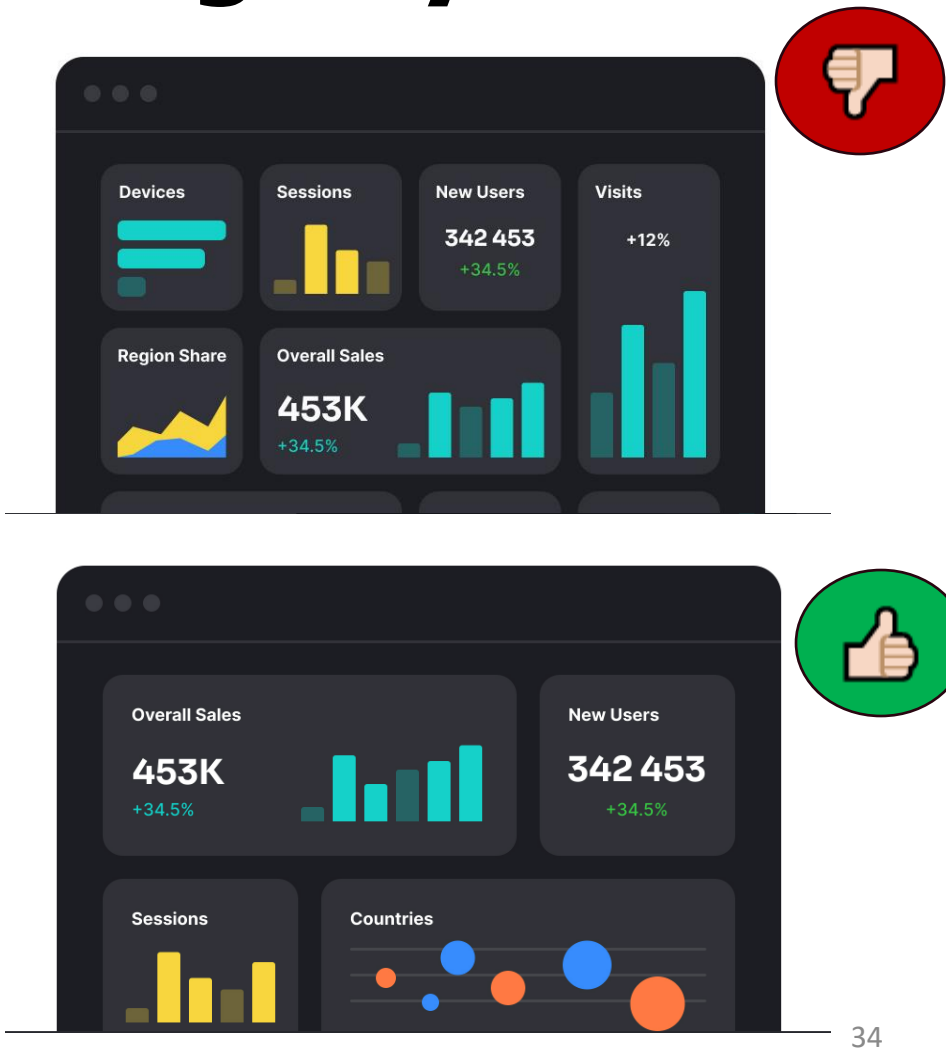


Common Issues with Poorly Designed Dashboards

1. Clutter and overwhelming layouts
2. Lack of hierarchical structure
3. Inconsistent design and misleading visuals
4. Neglecting user context and needs

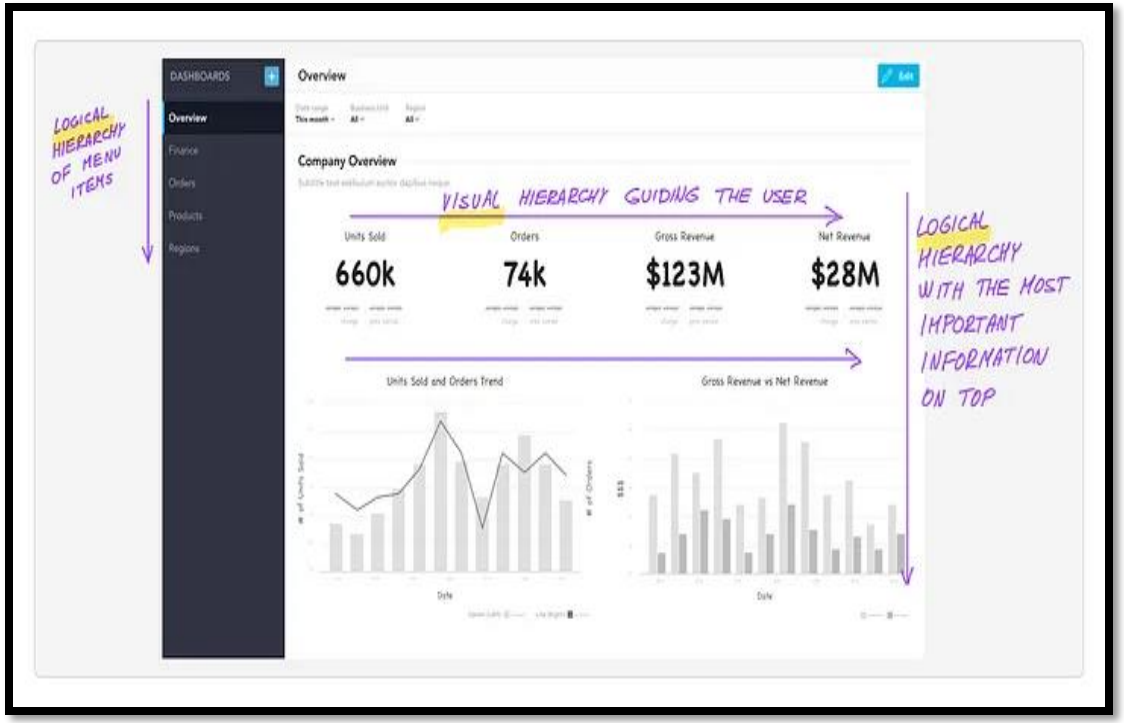
1. Clutter and Overwhelming Layouts

- Too many metrics and visualizations are crammed into one screen.
- Users struggle to identify what's most important.
- This cluttered layout can lead to confusion or cognitive load.



2. Lack of Hierarchical Structure

- Good hierarchy allows users to quickly identify the most important information.
- Guide the user's eye using:
  a. Size/ Typography
  b. Color
  c. Placement/ Layout
- Put the most important data top-left
- Group related content together



3. Inconsistent Design and Misleading Visuals

- Align elements neatly
- Consistent colors, fonts, and spacing
- Standardize KPI cards and filters
- Use grid layout for balance and symmetry



4. Neglecting User Context and Needs

- A one-size-fits-all dashboard rarely works.
- Effective dashboards should be tailored to the specific needs and context of the user.
- Example:
  a. Sales manager looking for revenue trends
  b. Product manager tracking feature usage

TAKE HOME ACTIVITY



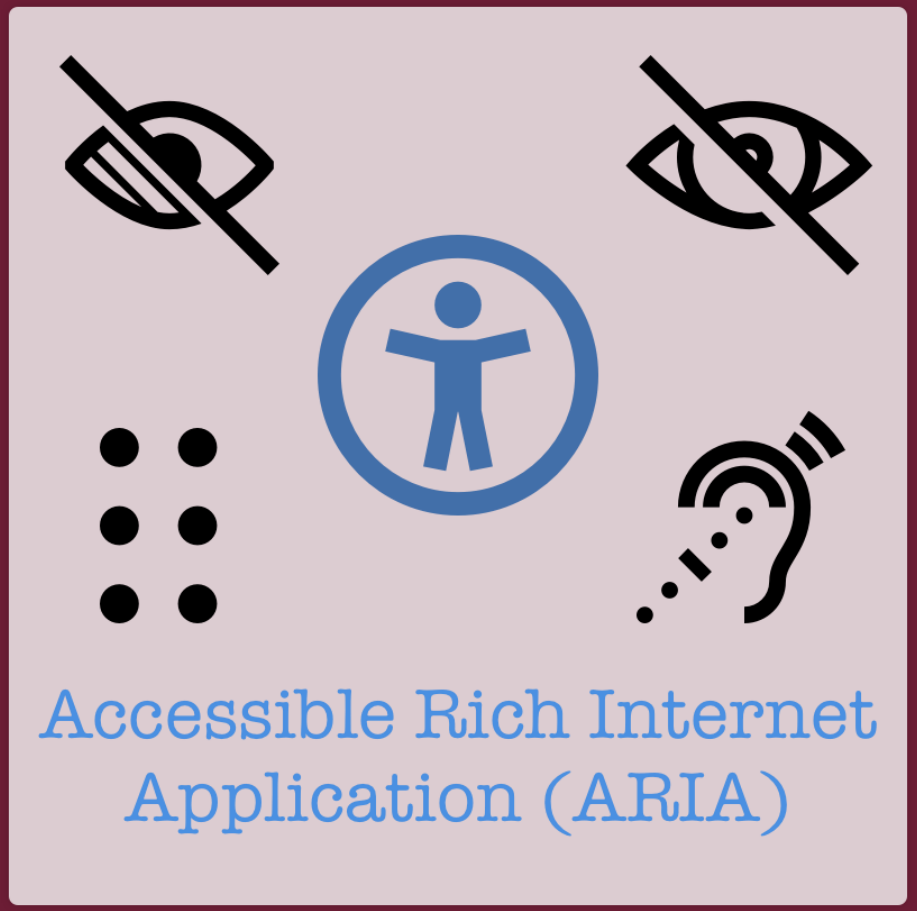
7 deadly SINS of data visualization

https://medium.com/design-bridges/dataviz-sins-976f3a08948c

Incorporate Accessibility Standards

1. Screen Reader Compatibility
    - Accessible Rich Internet Application:
        a. Visual – colorblind
        b. Hearing
        c. Motor - Inability to use a mouse, slow response time
        d. Cognitive – Learning disabilities, distractibility, inability to remember (elderly)
    - Ensure contrast and readable fonts
    - Provide textual descriptions for visuals
    - Avoid relying solely on color to convey meaning

2. Keyboard Navigation – all interactive elements are accessible via keyboard navigation



THANK YOU



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